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Android Sending and Receiving data from remote server using IntentService

September 26, 2013

By George Mathew

In this article, we will see how to develop an Android application that sends and receives data from a web server using IntentService. IntentService provides a simple mechanism to execute long lasting operations (eg : downloading data) in a background thread.

In this application user is able to select a country from a spinner widget and on selecting the country, corresponding capital name will be fetched from an http web server using IntentService and displayed to the user.



Major steps involved in this application are as follows :

- Populate Spinner widget with a String ArrayAdapter
- Create an IntentService class (CapitalService), which will be invoked when user selects a country from the Spinner widget
- Override the method “onHandleIntent()” of the IntentService class to fetch the capital of the selected country from a remote server
- Define a BroadcastReceiver class (CapitalReceiver) in MainActivity, to receive the broadcast message send from the IntentService class
- Override the method “onReceive()” of the BroadcastReceiver class to receive data from service and set in TextView of the MainActivity
- Register the broadcast receiver in MainActivity with Android’s LocalBroadcastManager

This application is developed in Eclipse 4.2.0 with Android SDK (22.2.1) and ADT plugin (22.2.1)

---

1. Create an Android application project with the given below project details

- Application Name : IntentServiceDemo
- Project Name : IntentServiceDemo

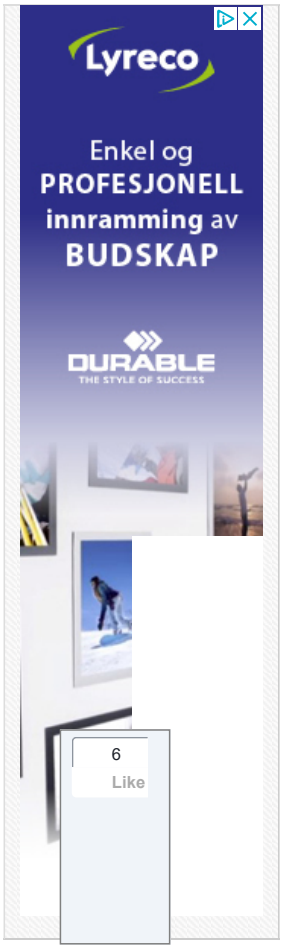
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- Package Name : in.wptrafficanalyzer.intentservicedemo
- Minimum Required SDK : API 8 : Android 2.2 (Froyo)
- Target SDK : API 18 : Android 4.3
- Compile With : API 18 : Android 4.3
- Theme : Holo Light with Dark Action Bar

---

2. Update the layout file res/layout/activity_main.xml for the MainActivity

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/an
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:paddingBottom="@dimen/activity_vertical_margin"
    android:paddingLeft="@dimen/activity_horizontal_margin"
    android:paddingRight="@dimen/activity_horizontal_margin"
    android:paddingTop="@dimen/activity_vertical_margin"
    tools:context=".MainActivity" >

    <Spinner
        android:id="@+id/spr_country"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerHorizontal="true" />

    <TextView
        android:id="@+id/tv_capital"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@id/spr_country"
        android:layout_centerHorizontal="true" />

</RelativeLayout>
```

---

3. Creating a layout for the Spinner Items in the file res/layout/spinner_layout.xml

```
<?xml version="1.0" encoding="utf-8"?></pre>
<TextView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:id="@+id/tv_item"
    android:padding="15dp"
    android:textColor="@android:color/black" >

</TextView>
```

---

4. Creating a class called Constants to define application constants in the file src/in/wptrafficanalyzer/intentservicedemo/Constants.java

```
package in.wptrafficanalyzer.intentservicedemo;

public final class Constants {

    // Action Name to IntentFilter
    public static final String BROADCAST_ACTION = "in.wptrafficanaly

    public static final String EXTRA_COUNTRY = "country";

    public static final String EXTRA_CAPITAL = "capital";

    // The url to the server where php script for capital is availab
    public static final String SERVER_URL = "http://wptrafficanalyze

}
```

---

5. Creating an IntentService class called CapitalService in the file src/in/wptrafficanalyzer/intentservicedemo/CapitalService.java

6
Like

```
1  package in.wptrafficanalyzer.intentservicedemo;
2
3  import java.io.BufferedReader;
4  import java.io.IOException;
5  import java.io.InputStream;
6  import java.io.InputStreamReader;
7  import java.io.UnsupportedEncodingException;
8  import java.net.HttpURLConnection;
9  import java.net.URL;
10 import java.net.URLEncoder;
11
12 import android.app.IntentService;
13 import android.content.Intent;
14 import android.support.v4.content.LocalBroadcastManager;
15 import android.util.Log;
16
17 public class CapitalService extends IntentService{
18
19     public CapitalService(){
20         super("Empty Constructor");
21     }
22
23     public CapitalService(String name) {
24         super(name);
25         // TODO Auto-generated constructor stub
26     }
27
28     // This method is executed in background when this service is s
29     @Override
30     protected void onHandleIntent(Intent intent) {
31         String country = "";
32         String capital = "";
33         String url = "";
34
35         try {
36
37             country = URLEncoder.encode(intent.getStringExtra(Const
38
39         } catch (UnsupportedEncodingException e1) {
40             // TODO Auto-generated catch block
41             e1.printStackTrace();
42         }
43
44         url = Constants.SERVER_URL + "/get_capital.php?country=" +
45
46         try {
47             // Retreiving capital from the url
48             capital = downloadUrl(url);
49
50             // Creating an intent for broadcastreceiver
51             Intent broadcastIntent = new Intent(Constants.BROADCAST
52
53             // Attaching data to the intent
54             broadcastIntent.putExtra(Constants.EXTRA_CAPITAL, capit
55
56             // Sending the broadcast
57             LocalBroadcastManager.getInstance(getApplicationContext
58
59         } catch (IOException e) {
60             // TODO Auto-generated catch block
61             e.printStackTrace();
62         }
63     }
64
65     /** A method to download data from url */
66     private String downloadUrl(String strUrl) throws IOException{
67         String data = "";
68         InputStream iStream = null;
69         HttpURLConnection urlConnection = null;
70         try{
71             URL url = new URL(strUrl);
72             // Creating an http connection to communicate with url
73             urlConnection = (HttpURLConnection) url.openConnection(
74
75             // Connecting to url
76             urlConnection.connect();
77
78             // Reading data from url
79             iStream = urlConnection.getInputStream();
```

```
80
81              bufferedReader br = new BufferedReader(new InputStreamR
82
83              StringBuffer sb = new StringBuffer();
84
85              String line = "";
86              while( ( line = br.readLine()) != null){
87                  sb.append(line);
88              }
89
90              data = sb.toString();
91
92              br.close();
93
94          }catch(Exception e){
95              Log.d("Exception while fetching data", e.toString());
96          }finally{
97              iStream.close();
98              urlConnection.disconnect();
99          }
100         return data;
101     }
102 }
```

---

6. Updating the class MainActivity in the file src/in/wptrafficanalyzer/intentservicedemo/MainActivity.java

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Like

```
1  package in.wptrafficanalyzer.intentservicedemo;
2
3  import android.app.Activity;
4  import android.content.BroadcastReceiver;
5  import android.content.Context;
6  import android.content.Intent;
7  import android.content.IntentFilter;
8  import android.os.Bundle;
9  import android.support.v4.content.LocalBroadcastManager;
10 import android.view.Menu;
11 import android.view.View;
12 import android.widget.AdapterView;
13 import android.widget.AdapterView.OnItemSelectedListener;
14 import android.widget.ArrayAdapter;
15 import android.widget.Spinner;
16 import android.widget.TextView;
17
18 public class MainActivity extends Activity {
19
20     // Array of countries
21     String[] mCountry = new String[] {
22             "India",
23             "Pakistan",
24             "Bangladesh",
25             "Srilanka",
26             "Nepal",
27             "China",
28             "Japan",
29             "South Korea",
30             "North Korea",
31             "Afghanistan"
32     };
33
34     Spinner mSprCountry;
35     TextView mTvCapital;
36     ArrayAdapter<String> mAdapter;
37     Intent mServiceIntent;
38     CapitalReceiver mReceiver;
39     IntentFilter mFilter;
40
41     @Override
42     protected void onCreate(Bundle savedInstanceState) {
43         super.onCreate(savedInstanceState);
44         setContentView(R.layout.activity_main);
45
46         // Adapter for Country Spinner
47         mAdapter = new ArrayAdapter<String>(getApplicationContext()
48
49         // Getting reference to TextView
50         mTvCapital = (TextView) findViewById(R.id.tv_capital);
```

```
51
52          // Getting reference to Country Spinner
53          mSprCountry = (Spinner) findViewById(R.id.spr_country);
54
55          // Setting adapter for the Country Spinner
56          mSprCountry.setAdapter(mAdapter);
57
58          // Creating an intent service
59          mServiceIntent = new Intent(getApplicationContext(), Capita
60
61          mSprCountry.setOnItemSelectedListener(new OnItemSelectedLis
62
63              @Override
64              public void onItemSelected(AdapterView<?> arg0, View ar
65                  int arg2, long arg3) {
66                  if(arg1!=null){
67                      String country = ((TextView)arg1).getText().toS
68                      mServiceIntent.putExtra(Constants.EXTRA_COUNTRY
69
70                      // Starting the CapitalService to fetch the cap
71                      startService(mServiceIntent);
72                  }
73              }
74
75              @Override
76              public void onNothingSelected(AdapterView<?> arg0) {
77                  // TODO Auto-generated method stub
78              }
79          });
80
81          // Instantiating BroadcastReceiver
82          mReceiver = new CapitalReceiver();
83
84          // Creating an IntentFilter with action
85          mFilter = new IntentFilter(Constants.BROADCAST_ACTION);
86
87           // Registering BroadcastReceiver with this activity for th
88           LocalBroadcastManager.getInstance(getApplicationContext())
89
90      }
91
92      @Override
93      public boolean onCreateOptionsMenu(Menu menu) {
94          // Inflate the menu; this adds items to the action bar if i
95          getMenuInflater().inflate(R.menu.main, menu);
96          return true;
97      }
98
99      // Defining a BroadcastReceiver
100     private class CapitalReceiver extends BroadcastReceiver{
101         @Override
102         public void onReceive(Context context, Intent intent) {
103             String capital = intent.getStringExtra(Constants.EXTRA_
104             mTvCapital.setText("Capital : " + capital);
105         }
106     }
107 }
```

---

7. Create a php script file namely “get_capital.php” in a web server having php support

```
1   <?php
2       $country = "none";
3       $capital = array(
4           "India"=>"New Delhi",
5           "Pakistan"=>"Islamabad",
6           "Bangladesh"=>"Dhaka",
7           "Srilanka"=>"Colombo",
8           "Nepal"=>"Kathmandu",
9           "China"=>"Beijing",
10          "Japan"=>"Tokyo",
11          "South Korea"=>"Seoul",
12          "North Korea"=>"Pyongyang",
13          "Afghanistan"=>"Kabul",
14      );
15      if(isset($_GET['country'])){
16          $country = $_GET['country'];
17      }
```

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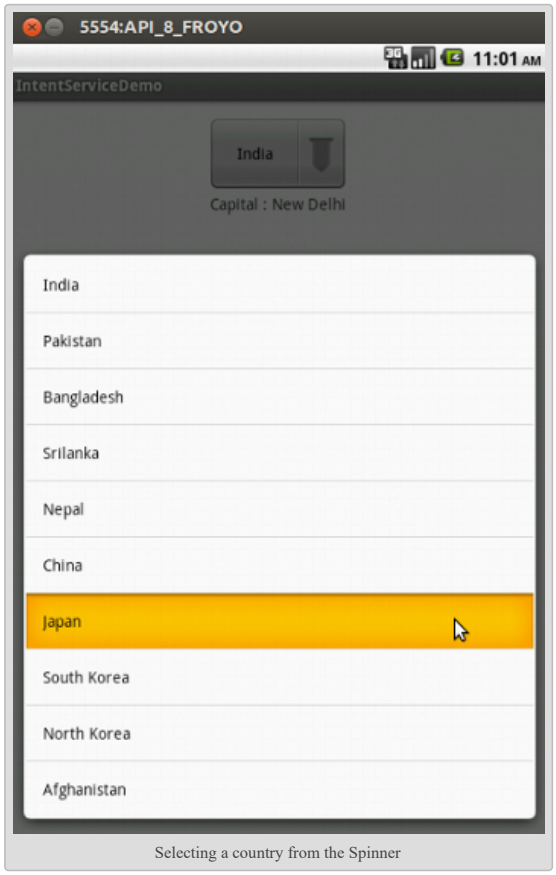
```
18
19      if(array_key_exists($country,$capital))
20          echo $capital[$country];
21      else
22          echo "Please pass a valid 'country' param from this list : "
```

Note : After creating this file, please ensure that, the variable ***SERVER_URL*** of the class ***Constants*** is updated with your server url.

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POWER Lille Grensen
Oslo 10:00–19:00

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8. Screenshot of the application



Selecting a country from the Spinner

9. Download the source code of the application

Android Sending and Receiving data from remote server using IntentService 515.71 KB

Download Full Source Code (http://wptrafficanalyzer.in/blog/?wpdmdl=9376)

How to hire me?



I am George Mathew, working as software architect and Android app developer at wptrafficanalyzer.in

You can hire me on hourly basis or on project basis for Android applications development.

For hiring me, please mail your requirements to info@wptrafficanalyzer.in.

My other blogs
store4js.blogspot.com

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Tags: BroadcastReceiver, IntentService

5 Responses to *Android Sending and Receiving data from remote server using IntentService*

PUNITH on February 12, 2014 at 4:12 pm

Your blog is awesome for android programming learner. I recommend this for everyone.

REPLY

Chaitanya on February 15, 2014 at 8:35 pm

Dear Mathew

Thank for your good site.I should develop an application in which it should be controlled from a remote place. Controlled in the sense, suppose I want to change the font or text color, I should be able to change it from a remote web server or web page. In the webpage I will have an option of selecting different fonts, sizes, color and so on. When I select particular font, size, color and so on, it should reflect in the application that I develop.

Could you please help me out in this case.

Please consider my request favorably. Awaiting for a positive reply.

REPLY

Chandan on December 2, 2014 at 4:30 pm

Dear Mathew,

Your code fetch the value based on Country select and display the capital name.

I want this in this way.
i want the Country and Capital name from Android app using PHP

Please help…
Thank you.

REPLY

edosat on June 6, 2015 at 5:32 pm

Question 1.
SERVER_URL = “http://wptrafficanalyzer.in/p/intent_service_demo”;
how and where i can get this path?
Question 2.
Where should i save “get_capital.php” how i save and use it?

REPLY

edosat on June 7, 2015 at 1:06 am

what argument i give to SERVER_URL =””?”, where should i save php script file namely “get_capital.php” in a web server having php support

REPLY

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